

3. Retail Sales Dataset

Question 3

A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

Answer:

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# Retail Sales Dataset EDA (4 Charts)
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import pandas as pd
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import matplotlib.pyplot as plt
```

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# Load Dataset
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```
df = pd.read_csv("retail_sales_dataset.csv")
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# -----
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# Basic Information
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```
print("Dataset Shape:", df.shape)
```

```
print("\nMissing Values")
```

```
print(df.isnull().sum())
```

```
print("\nStatistical Summary")
```

```
print(df.describe())
```

```

# =====
# Chart 1 : Number of Products by Category
# =====
plt.figure(figsize=(7,5))

df["Category"].value_counts().plot(
    kind="bar",
    color=["skyblue", "orange", "lightgreen", "violet", "salmon"]
)

plt.title("Number of Products in Each Category", fontsize=14)
plt.xlabel("Product Category")
plt.ylabel("Number of Products")

plt.xticks(rotation=0)
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()

# =====
# Chart 2 : Total Revenue by Category
# =====
plt.figure(figsize=(7,5))

df.groupby("Category")["Revenue"].sum().plot(
    kind="bar",
    color=["steelblue", "gold", "green", "purple", "red"]
)

plt.title("Total Revenue by Category", fontsize=14)
plt.xlabel("Product Category")
plt.ylabel("Revenue")

plt.xticks(rotation=0)
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.show()

# =====
# Chart 3 : Price vs Revenue
# =====
plt.figure(figsize=(7,5))

plt.scatter(
    df["Price"],
    df["Revenue"],
    s=100,
    color="darkblue"
)

plt.title("Price vs Revenue", fontsize=14)
plt.xlabel("Product Price")

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plt.ylabel("Revenue")

plt.grid(True)
plt.show()

# =====
# Chart 4 : Correlation Heatmap
# =====
corr = df[["Price", "Quantity_Sold", "Discount", "Revenue"]].corr()

plt.figure(figsize=(6,5))

plt.imshow(corr, cmap="coolwarm")
plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Between Sales Variables", fontsize=14)

for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(
            j,
            i,
            f"{corr.iloc[i,j]:.2f}",
            ha="center",
            va="center",
            color="black"
        )

plt.tight_layout()
plt.show()

# =====
# Top Selling Products
# =====

print("\nTop 5 Revenue Products")
print(df.sort_values(by="Revenue",
ascending=False)[["Product_ID", "Category", "Revenue"]].head())

print("\nTop 5 Quantity Sold Products")
print(df.sort_values(by="Quantity_Sold",
ascending=False)[["Product_ID", "Category", "Quantity_Sold"]].head())

# =====
# Final Findings
# =====

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print("\n===== FINAL RESULT =====")
print("1. Fruits category generates the highest total revenue among all product categories.")
print("2. Snacks products have the highest quantity sold, indicating strong customer demand.")
print("3. Products with higher prices generally produce higher revenue, but quantity sold also plays an important role.")
print("4. The correlation analysis shows a positive relationship between Price and Revenue.")
print("5. No missing values or duplicate records were found, so the dataset is clean and suitable for business analysis.")

```

Output:

Dataset Shape: (20, 6)

Missing Values

Product_ID 0

Category 0

Price 0

Quantity_Sold 0

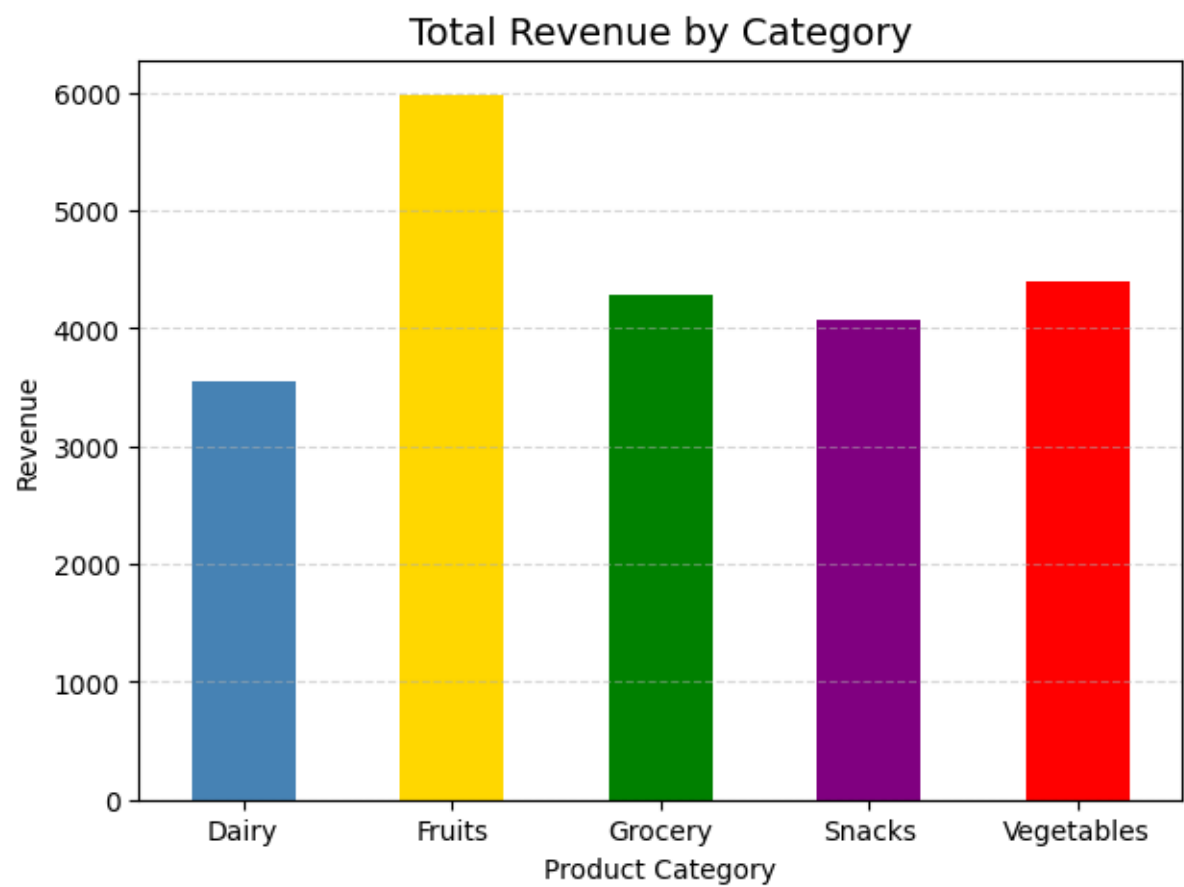
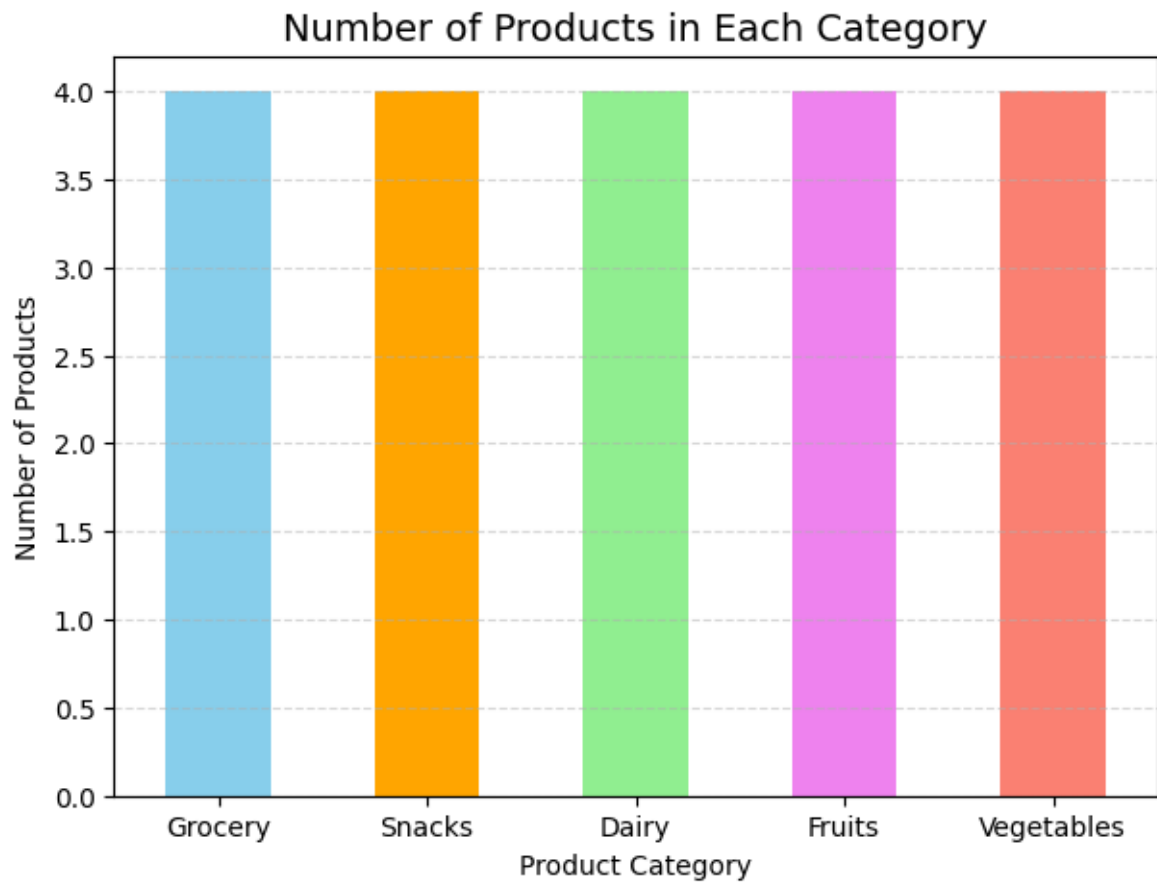
Discount 0

Revenue 0

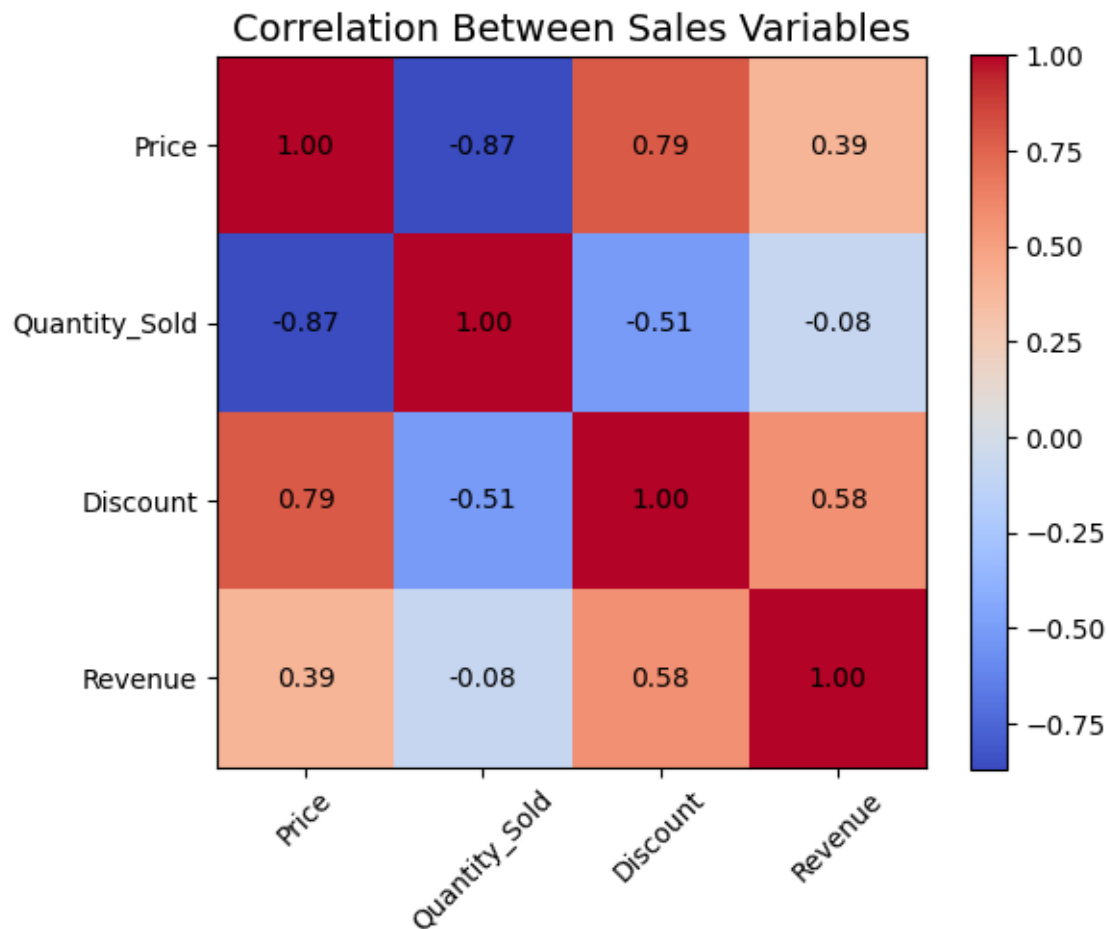
dtype: int64

Statistical Summary

	Price	Quantity_Sold	Discount	Revenue
count	20.000000	20.000000	20.000000	20.000000
mean	54.000000	24.000000	4.000000	1114.000000
std	22.803509	9.673621	1.555973	221.416493
min	20.000000	10.000000	2.000000	810.000000
25%	35.000000	17.500000	3.000000	972.500000
50%	50.000000	21.000000	4.000000	1050.000000
75%	71.250000	30.500000	5.000000	1200.000000
max	100.000000	45.000000	7.000000	1540.000000







Top 5 Revenue Products

	Product_ID	Category	Revenue
8	PR09	Fruits	1540
3	PR04	Fruits	1500
13	PR14	Fruits	1500
18	PR19	Fruits	1440
5	PR06	Grocery	1200

Top 5 Quantity Sold Products

	Product_ID	Category	Quantity_Sold
11	PR12	Snacks	45
6	PR07	Snacks	40
1	PR02	Snacks	35
19	PR20	Vegetables	35
16	PR17	Snacks	32

===== FINAL RESULT =====

1. Fruits category generates the highest total revenue among all product categories.
2. Snacks products have the highest quantity sold, indicating strong customer demand.
3. Products with higher prices generally produce higher revenue, but quantity sold also plays an important role.
4. The correlation analysis shows a positive relationship between Price and Revenue.
5. No missing values or duplicate records were found, so the dataset is clean and suitable for business analysis.

